Computer Science & IT



Data Structure

Tree

Lecture No. 05



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Recap of Previous Lecture



- Topic — Traversal
- Topic — preorder
- Topic — Inorder
- Topic — postorder
- Topic — Construction of tree < preorder : Inorder / postorder Inorder

Slide

Topics to be Covered



Topic

Topic

Topic

Topic

Topic

Binary Search Tree

Basic Search tree basic operation

Tree construction

Binary Search tree traversal

Slide



Topic : Tree

BST
BST Preorder Post Order
BST Search , MIN MAX
BST DELETE
Counting
Problem Solving

Slide



Topic : Binary Search Tree

Binary Search tree in which all key value (Label of node) less than parent node goes into Left subtree & all key value greater than parent node goes to right subtree. This property is True for subtrees as well.



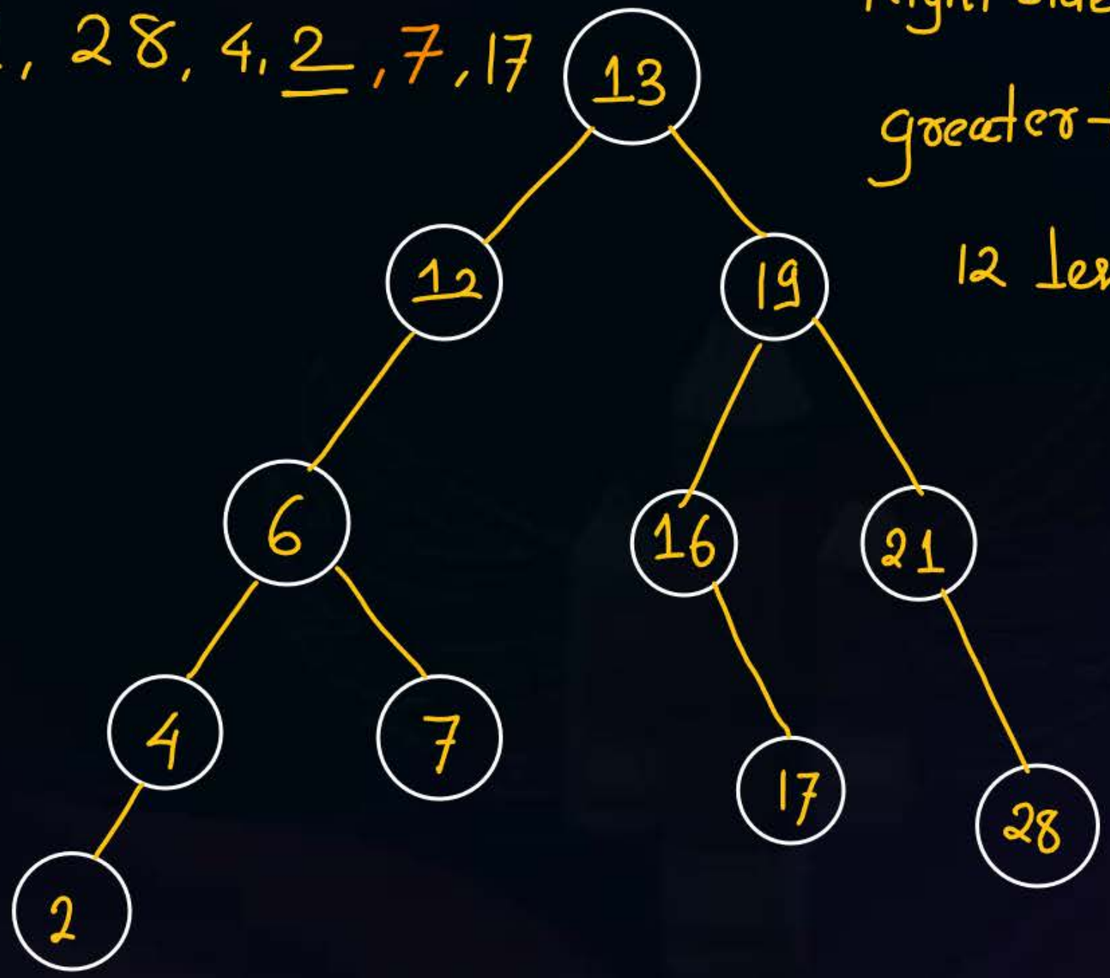
Topic : Binary Search Tree

Construct the BST by inserting following key values

13, 12, 6, 19, 16, 21, 28, 4, 2, 7, 17

Right side greater than 12 lesser than B

Binay Search tree Height



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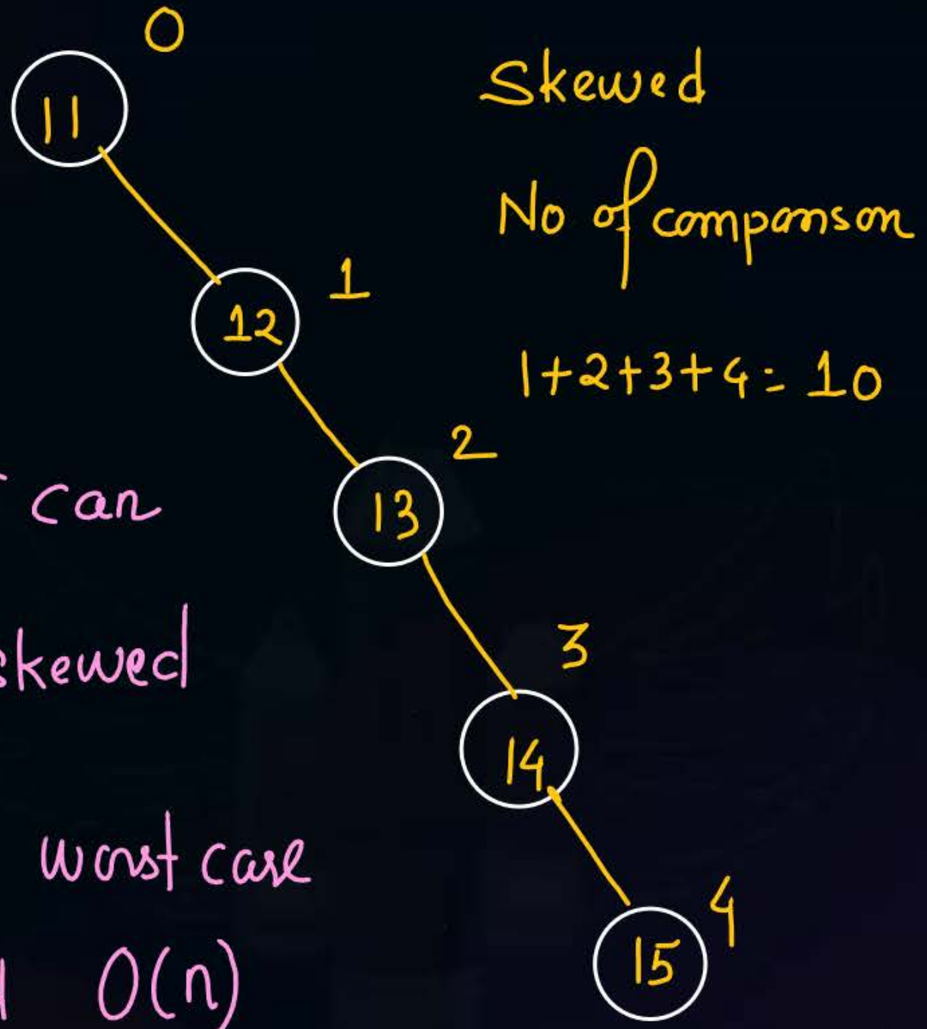
Topic : Binary Search Tree

Binary Search Tree Height

#Q if 11, 12, 13, 14, 15 key are inserted in empty BST in the given order then No. of comparison required to construct the tree is ___

BST can be skewed

BST worst case height O(n)



Skewed

No of comparison

1+2+3+4 = 10

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Question

#Q Create Binary Search Tree
40, 20, 60, 50, 30, 10, 70
Number of leaves in the generated tree is_____.



Minimum Height tree with

7 Nodes

Slide



Topic : Question

Q A binary search tree is generated by inserting in order the following integers:

50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24

number of node in left and right subtree us

(A) [4, 7]

(B) [7, 4]

(C) [8, 3]

(D) [3, 8]

7 Nodes , 4 Node

Left right



Topic : Binary Search Tree

#Q The binary search tree shown below was constructed by inserting a sequence of items into an empty tree



Which of the following input sequences will produce this binary search tree?

(a) 5,3,4,9,12,7,8,6,20

(b) 5,93,768,4,12,20

(c) 5,9,738,12,6,420

(d) 5,936,78,4,1220

Slide



Topic : Binary Search Tree

#Q The binary search tree shown below was constructed by inserting a sequence of items into an empty tree



Which of the following input sequences will produce this binary search tree?

(a) 5,3,4,9,12,7,8,6,20

(b) 5,9,3,7,6,8,4,12,20

(c) 5,9,7,3,8,12,6,4,20

(d) 5,9,3,6,7,8,4,12,20

Slide



Topic : Binary Search Tree

#Q The binary search tree shown below was constructed by inserting a sequence of items into an empty tree



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Slide



Topic : Binary Search Tree

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(a) 5,3,4,9,12,7,8,6,20

(b) 5,9,3,7,6,8,4,12,20

(c) 5,9,7,3,8,12,6,4,20

(d) 5,9,3,6,7,8,4,12,20 ✗

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Topic : Binary Search Tree

Traversal of BST

preorder : Root

13, 12, 6, 4, 2, 7, 19, 16, 17, 21, 28



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Topic : Binary Search Tree

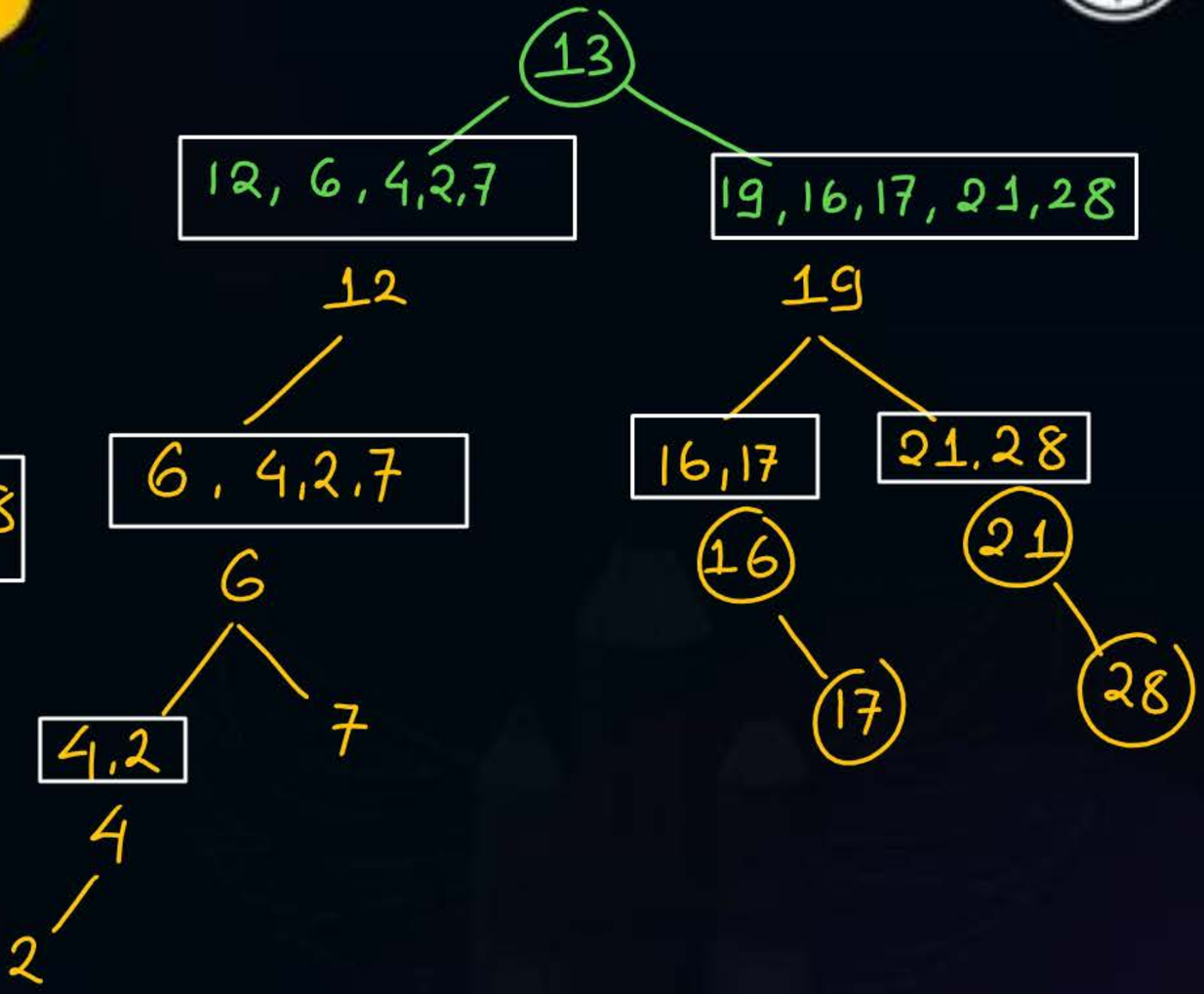
Traversal of BST

preorder : Root

13, 12, 6, 4, 2, 7, 19, 16, 17, 21, 28

Root — Left subtree — right subtree

only preorder we can construct BST



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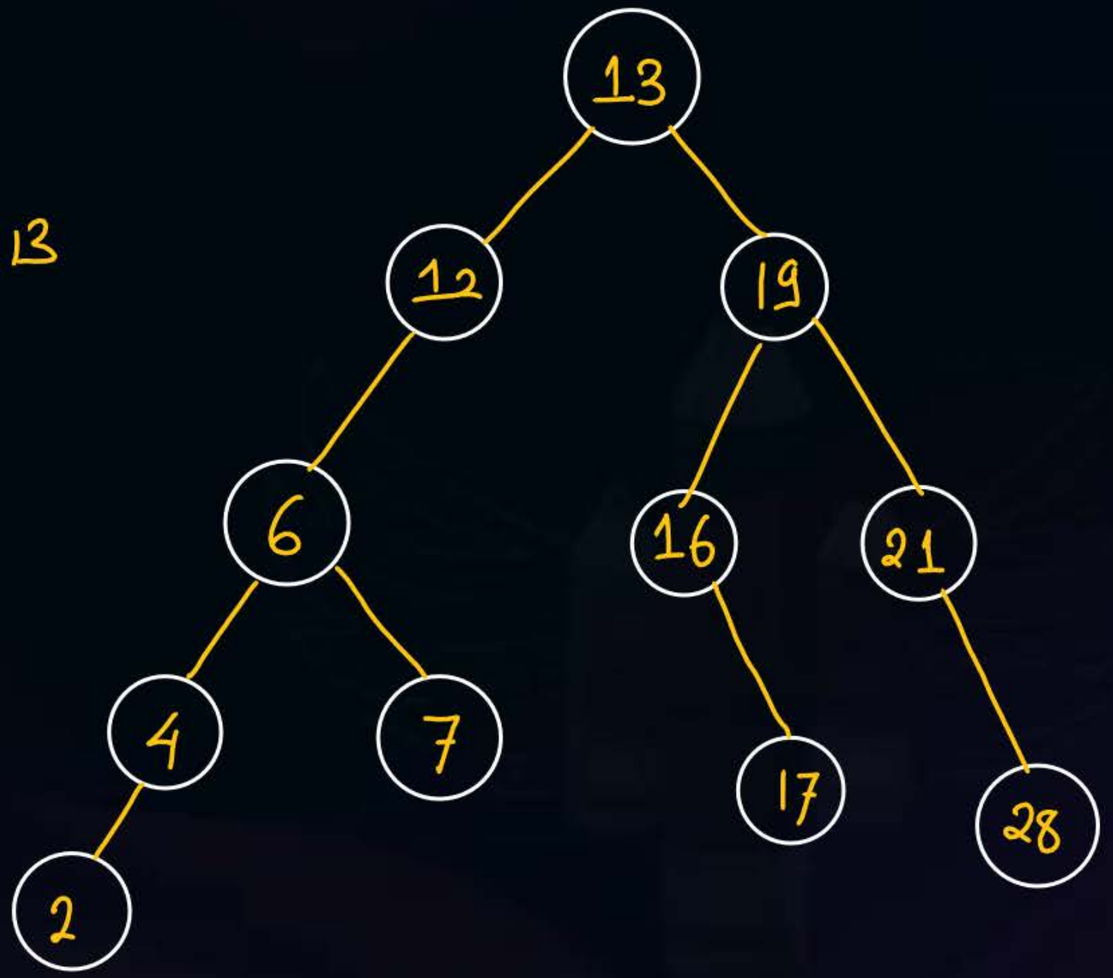


Topic : Binary Search Tree

Traversal of BST

postorder

2, 4, 7, 6, 12, 17, 16, 28, 21, 19, 13



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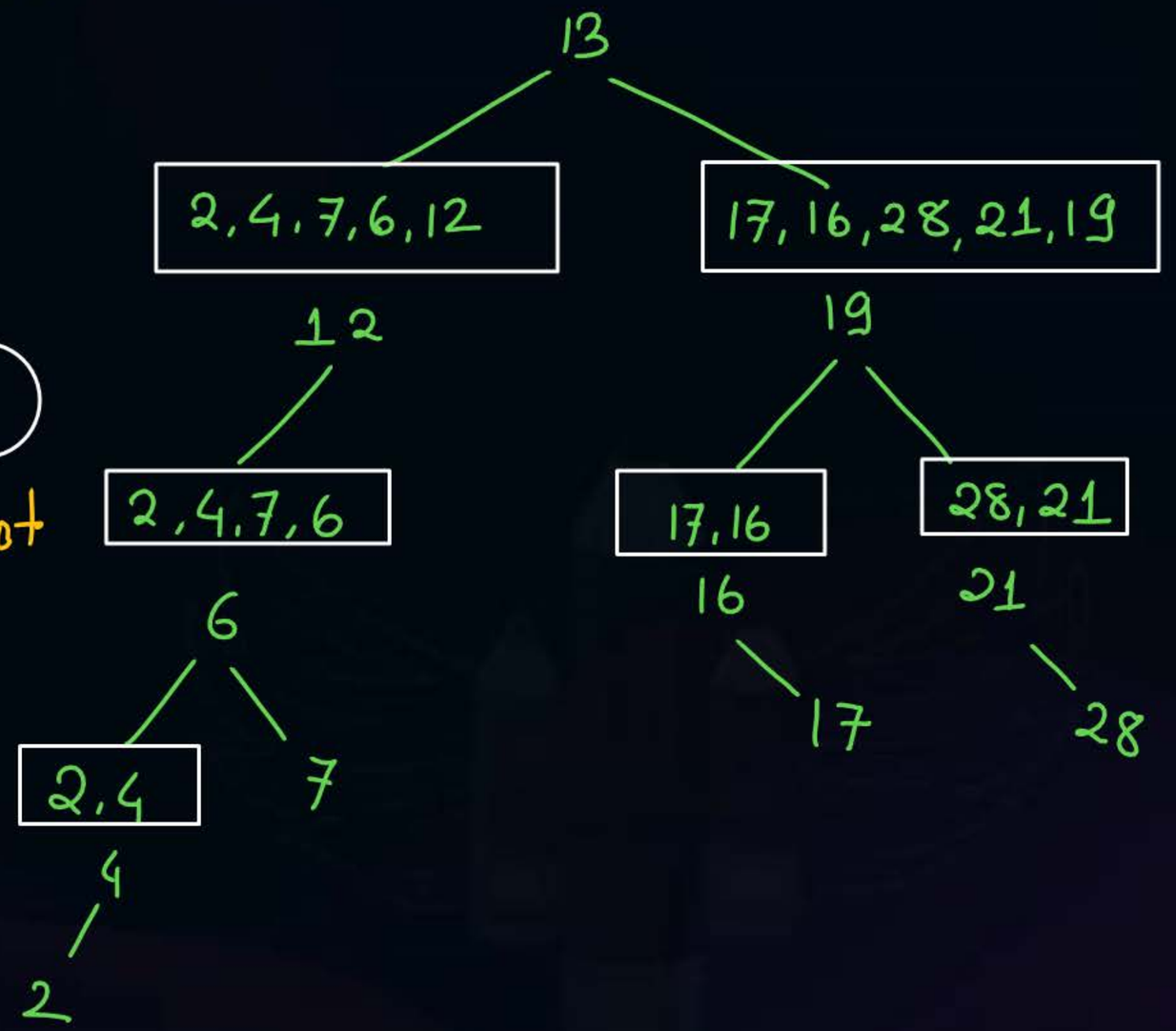
Topic : Binary Search Tree

Traversal of BST

postorder

2, 4, 7, 6, 12	17, 16, 28, 21, 19	13
Left	right	Root

Complete the tree



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Topic : Binary Search Tree

Traversal of BST

Searching 7



Inorder

2, 4, 6, 7, 12, 13, 16, 17, 19, 21, 28

Inorder sequence is Increasing order Sequence

Can we identify root element ??

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Topic : Binary Search Tree

MSQ

The Preorder traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, 42.

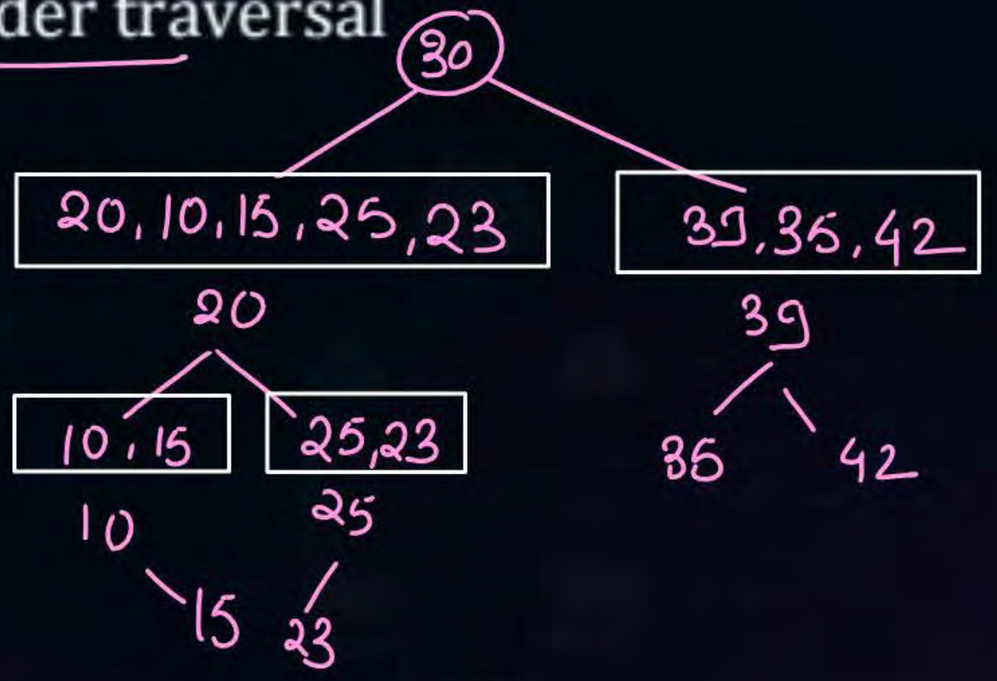
Which one of the following is the not Post order traversal sequence of the same tree?

(a) 10, 20, 15, 23,25, 35, 42, 39,30 X

(b) 15, 10, 25, 23, 20, 42, 35, 39, 30 X

(c) 15, 20, 10, 23, 25, 42, 35, 39, 30 X

(d) 15, 10, 23, 25,20, 35, 42, 39,30 postorder



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Topic : Binary Search Tree

```
int BSTsearch(Tnode* t, int x) {
    if(t) {
        if(t->data==x)
            return 1;
        else if(t->data>x)
            return BSTsearch(t->left, x);
        else
            return BSTsearch(t->right, x);
    }
    else return -1;
}
```



x = 15

Time Complexity

Search =

O(height) — BST - O(n)

Balanced BST - O(log n)

Slide



Topic : Question

#Q. You are given a set V of distinct integers. A binary search tree T is created by inserting all elements of V one by one, starting with an empty tree. The tree T follows the convention that, at each node, all values stored in the left subtree of the node are smaller than the value stored at the node. You are not aware of the sequence in which these values were inserted into T. and you do not have access to T.

Which one of the following statements is TRUE?

Set unordered Collection

(A) Postorder traversal of T can be determined from V ✗

(B) Inorder traversal of T can be determined from V ✓ Increasing order

(C) Root node of T can be determined from V ✗

(D) Preorder traversal of T can be determined from V ✗

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Topic : Binary Search Tree

Is it mandatory that min & max is a leaf Node

rightmost Node is Max

Leftmost Node is Min

```
int min (Tnode *t) {
    if(t) {
        while (t->left != NULL)
            t = t->left;
        return t->data;
    }
    printf("Empty tree");
} return -1;
```

O(h) — BST O(n)

balanced BST O(log₂n)



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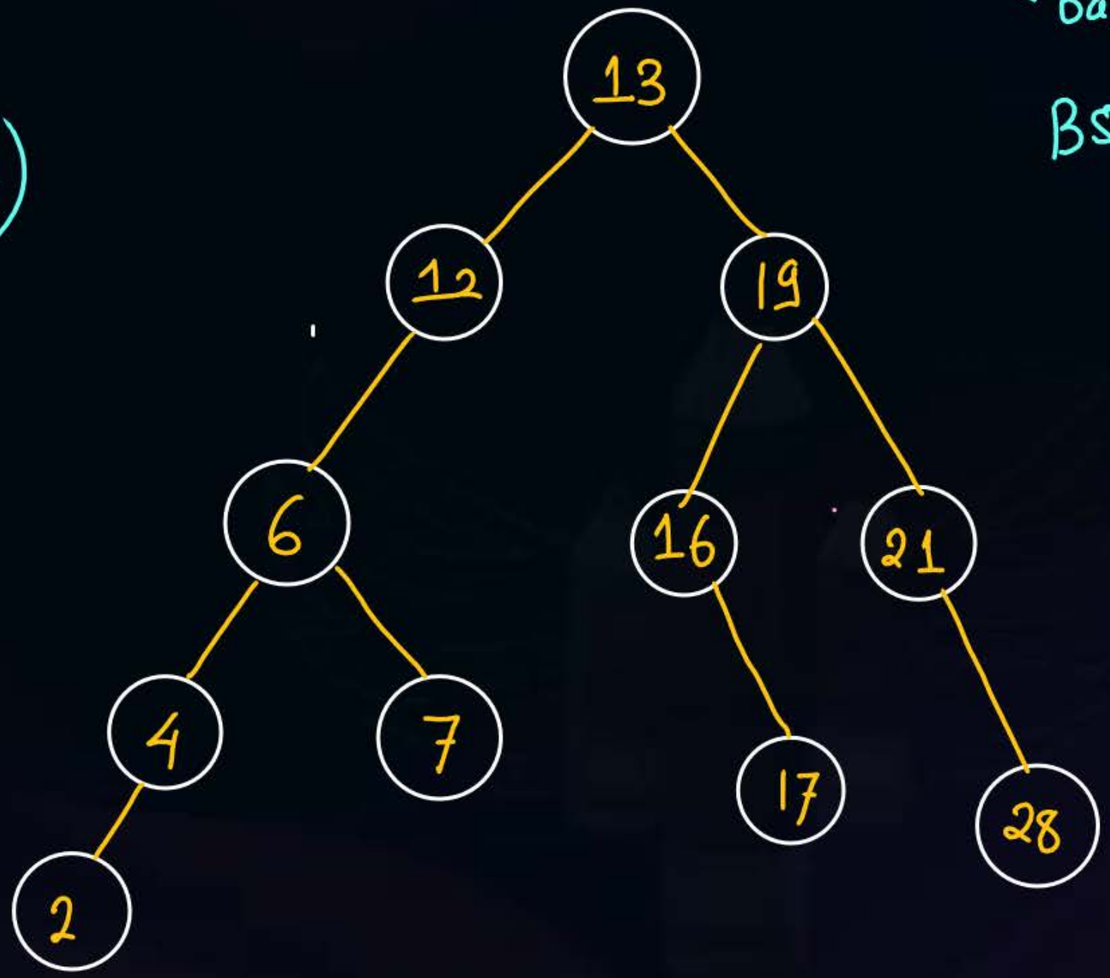


Topic : Binary Search Tree

```
int max(Tnode* t) {
    if(t) {
        while (t->right != NULL)
            t = t->right;
        return t->data;
    }
    else {
        printf("Empty tree");
        return -1;
    }
}
```

O(height) — BST : O(n), balanced

BST : O(log₂n)



Slide



Topic : Binary Search Tree

#Q Suppse BST constructed using Sequential representation with minimum Size of array with 33 elements. if Index of Smallest element is x and index of second largest elemement is y then x+y is 47

Index start from 1.

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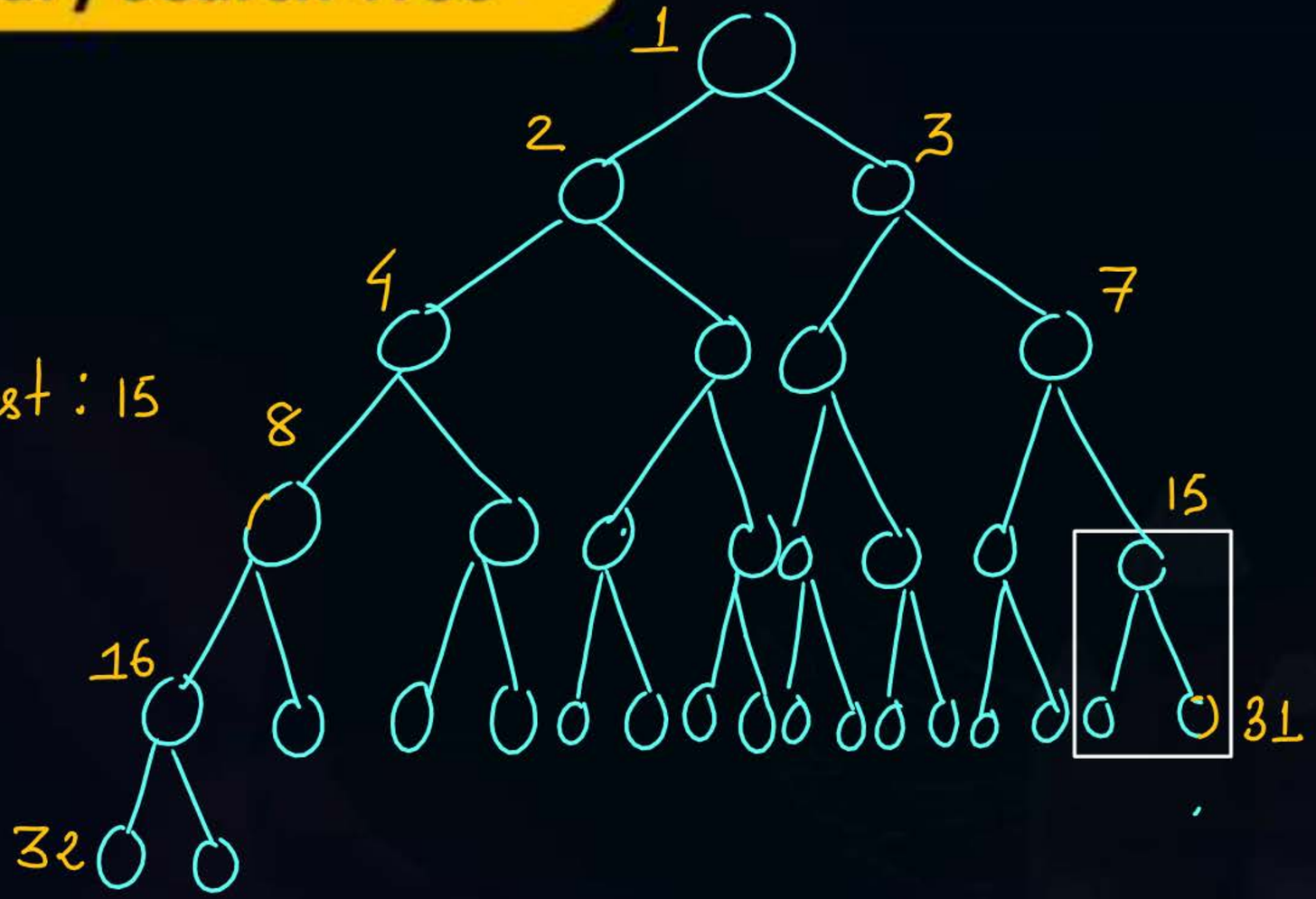


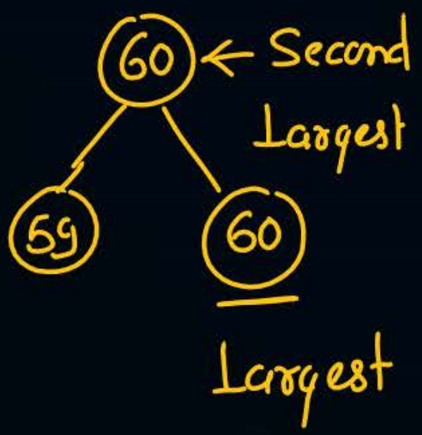
Topic : Binary Search Tree

X = 32 Smallest

Y = Second Largest : 15

32 + 15 = 47



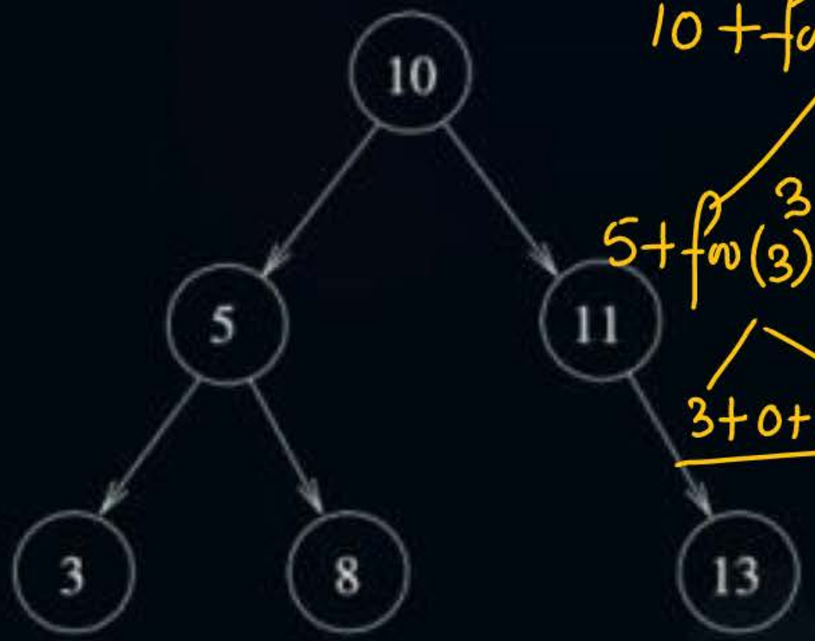


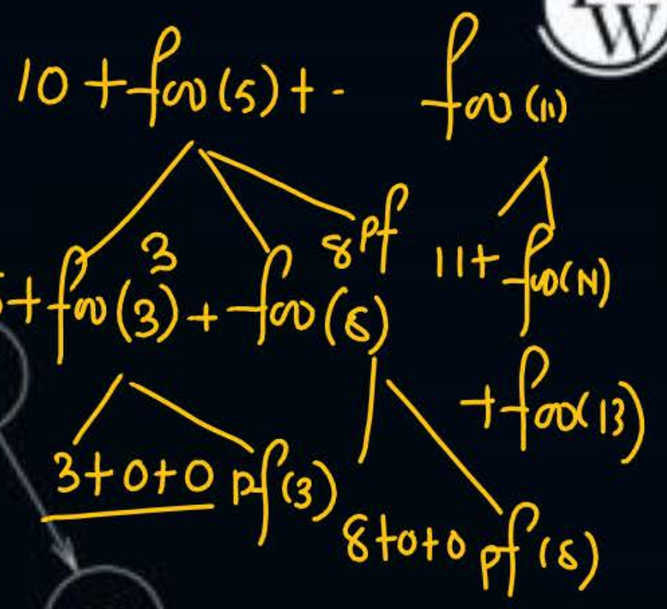
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Topic : Question

Consider the function and the binary tree shown.

```
int foo(node *p) {
    int retval;
    if (p == NULL)
        return 0;
    else {
        retval = p->val + foo(p->left) + foo(p->right);
        printf("%d ", retval);
        return retval;
    }
}
```





A. 3 8 5 13 11 10

B. 3 5 8 10 11 13

C. 3 8 16 13 24 50

D. 3 16 8 50 24 13

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2 mins Summary

Topic — Binary Search tree

Topic — Construction of BST

Topic — Traversal of BST

Topic — Search Min, Max

Topic

Slide

THANK - YOU

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